Rule: distribute dot over add

```
(d: dot (add A B) C)
=> (add (left: dot A C) (right: dot B C))
```

Reusable operation semantics

```
add: Add(X,Y)
dot: Reduce(K,Product(X,Y))
```

1. Axes IR

Source

```
Reduce(K,
  Product(Add(A,B),C))
```

Contract K; output I ++ J.

RHS template

```
Add(
  Reduce(K1,Product(A,C)),
  Reduce(K2,Product(B,C)))
```

K_1, K_2 unresolved

Reads retain coordinates.

2. Normalized terms

Source

```
[(1,[A,C],K),
 (1,[B,C],K)]
```

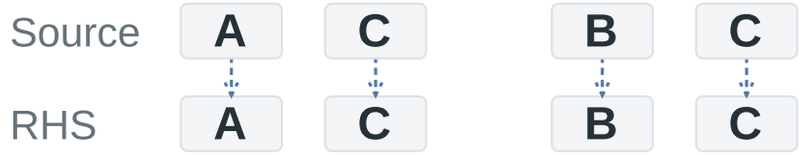
RHS arithmetic skeleton

```
[(1,[A,C],?),
 (1,[B,C],?)]
```

Term = (weight, reads, axes)
I_T: ordered axes of tensor T.
?: unresolved axis information

3. Attribute matching

Match operand reads



Reuse source coordinates

```
A,B: [I,K]; C: [K,J]
```

Construct attributes

Contract K; preserve I ++ J.

```
A → pos(I_A,K)
B → pos(I_B,K)
C → pos(I_C,K)
```

Constructed replacement

```
(add (left: dot A C) (right: dot B C))
```

`left.attrs=d.attrs; right.attrs=d.attrs` Separate holes

Verify completed rule

Legality, values, output order
No broadcast; distribution allowed.